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Supplemental Material

Quantification of 16 Metals in Fluids and Aerosols From Ultrasonic Pod-Style Cigarettes and Comparison to Electronic Cigarettes

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Figure S1. Scanning electron microscopy (SEM) and energy dispersive X-ray spectrometry (EDS) of SURGE components.

Figure S2. EDS spectra of SURGE sonicator components. (A) vaped cotton wick, which was mainly carbon, oxygen, silicon and lead. (B) sonicator spring, which was mainly nickel, and gold.

Figure S3. Concentrations of Elements/metals in Aerosols Across Groups. (A) Silicon, (B) Aluminum, (C) Lead, (D) Zinc, (E) Copper, (F) Chromium, (G) Tin, (H) Iron, (I), Silver, (J) Cobalt, (K), Cadmium, (L) Manganese. The x-axis indicates the groups and the y-axis shows concentrations of elements in µg/L. Each bar is the mean +/- standard deviation of samples in each group. For comparisons of two groups, an unpaired t-test was used. When comparisons were done for three groups, a one-way ANOVA was used. * = $p < 0.05$; ** = $p < 0.01$; *** = $p < 0.001$; **** = $p < 0.0001$. Brackets are comparisons between groups where indicated.

Table S1: Elements, elemental class, limits of quantification, and limits of detection

Element	Elemental Class	Limit of Quantification (mg/L)	Limit of Detection (mg/L)
Aluminum	Post-Transition/Basic	0.01	0.0043
Arsenic	Metalloid	0.01	0.0077
Cadmium	Transition	0.001	0.00069
Chromium	Transition	0.003	0.002
Cobalt	Transition	0.003	0.00066
Copper	Transition	0.003	0.0024
Iron	Transition	0.01	0.00066
Lead	Post-Transition/Basic	0.01	0.0067
Manganese	Transition	0.0005	0.000086
Nickel	Transition	0.001	0.00081
Selenium	Metalloid	0.03	0.0166
Silicon	Metalloid	0.01	0.0027
Silver	Transition	0.003	0.0023
Tin	Post-Transition/Basic	0.01	0.004
Titanium	Transition	0.001	0.00035
Zinc	Transition	0.01	0.0005

Table S2: wt % of Elements in SURGE U-Cigarette Components

Element	Cotton (UV)	Cotton (V)	Ceramic (UV)	Ceramic (V)	Seal Assembly (Shell & Top)	Seal Block Bottom	Spring	Sensor
C	99.29	50.41	7.81	49.04	11.86	43.17	32.16	49.8
O	0	46.82	54.11	45.01	28.49	13.8	6.51	27.74
Si	0.01	0.27	29.48	4.4	11.24	2.89	5.16	3.5
Ni	0	0	0.11	0.02	9.89	26.09	31.06	0.22
Pb	0.38	1.34	0.04	0.05	22.45	0.28	0.1	0.22
Au	0	0	0	0	5.05	6.29	15.43	0.04
Sn	0	0	0.18	0.05	0	1.23	0	13.42
Fe	0	0.01	0.2	0.03	0	0.06	5.5	0.01
Cu	0	0.28	0.09	0.33	3.44	1.01	1.25	0.64
Al	0	0.02	1.78	0.11	3.12	0.22	0.08	0.39
Ag	0	0	0.06	0.03	2.68	0.05	0.06	0.27
P	0	0	2.08	0.13	0.04	2.07	0.02	0.04
K	0	0.01	1.71	0.11	0.04	0.03	0	0
Ba	0	0.01	0	0.01	0	1.31	0	1.68
Zn	0	0	0.1	0.11	1.45	0.09	0.13	0.37
Cr	0	0	0.01	0.01	0	0.03	1.52	0
Na	0	0.09	1.5	0.1	0.04	0.12	0.1	0.1
Mo	0.28	0	0.29	0	0	0.28	0.64	0
S	0	0.11	0	0.03	0.01	0.49	0	0.57
Cl	0.02	0.53	0.24	0.34	0	0.06	0	0.11
Ti	0	0	0.05	0.01	0.06	0.16	0	0.35
Mg	0	0.1	0.02	0.07	0.03	0.1	0.04	0.27
Ca	0	0	0.07	0	0.08	0.17	0.06	0.19
Mn	0	0	0	0	0.04	0.02	0.09	0.05
Co	0	0	0.07	0	0	0	0.08	0

EDS detection limit ranges = 0.1 - 0.5 wt%

Table S3: Mean Concentration (µg/L) of Silicon in U-cigarettes and E-cigarettes

Silicon	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Green Mint	4536 (4436 - 4612)	4103 (3927 - 4346)	1914 (1435 - 2559)
Polar Mint	4685 (4080 - 5349)	4006 (3535 - 4429)	1142 (459 - 1585)
Blueberry Ice	3873 (3782 - 3951)	4507 (3941 - 4963)	1909 (1774 - 2142)
Watermelon Ice	5350 (4833 - 6077)	4911 (4486 - 5288)	1627 (711 - 2975)
Fruit Medley	1750 (1638 - 1832)	1343 (1088 - 1605)	498 (458 - 561)
Mango	2754 (2584 - 2952)	2700 (2520 - 2888)	1734 (1617 - 1918)
Classic Menthol	3398 (2547 - 4142)	3691 (3619 - 3823)	883 (781 - 1020)
Classic Tobacco	5398 (5135 - 5590)	4165 (3958 - 4373)	1201 (451 - 1841)
Cool Mint	8078 (7921 - 8202)	6475 (5030 - 7927)	4778 (4083 - 6003)
Cool Cucumber	9344 (8490 - 10453)	12020 (11050 - 13683)	6322 (5365 - 7325)
Virginia Tobacco	25086 (12707 - 31483)	18619 (13209 - 25742)	10194 (5769 - 14965)
Crème Brulee	31327 (6660 - 47872)	19824 (4687 - 29623)	14104 (2690 - 22800)
SMOK Infinix (MB) ^a	-	136 (106 - 159)	576 (511 - 702)
SMOK Infinix (MM) ^b	-	199 (158 - 243)	300 (221 - 415)
Suorin Edge (MM) ^b	-	141 (98 - 165)	225 (150 - 350)
JUUL-Jones (CM) ^d	865 (832 - 922)	966 (825 - 1077)	681 (625 - 754)
PHIX (HS) ^d	1602 (1522 - 1657)	2165 (1894 - 2686)	2058 (1289 - 2546)
KILO 1K (DFI) ^e	9734 (9009 - 10853)	8755 (8309 - 9100)	6567 (6364 - 6713)
KWIT Stick (MT) ^f	3084 (2346 - 3927)	3788 (3433 - 3988)	3001 (2758 - 3170)

^a MB = Mango Bomb, ^b MM = Mentholate Mango, ^c CM = Clear Mango,

^d HS = Hard Strawberry, ^e DFI = Dewberry Fruit Ice, ^f MT = Mango Tango.

Columns with “ - ” = no data.

Table S4: Mean Concentration (µg/L) of Nickel in U-cigarettes and E-cigarettes

Nickel	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Green Mint	17 (8 - 27)	12 (10 - 14)	1 (-)
Polar Mint	26 (6 - 43)	29 (21 - 37)	-
Blueberry Ice	14 (7 - 20)	24 (16 - 36)	1 (-)
Watermelon Ice	16 (8 - 23)	41 (15 - 70)	-
Fruit Medley	1 (-)	29 (24 - 34)	6 (4 - 7)
Mango	1 (0 - 1)	30 (26 - 33)	1 (1 - 2)
Classic Menthol	8 (6 - 11)	19 (9 - 34)	-
Classic Tobacco	-	2 (1 - 3)	-
Cool Mint	1 (1 - 1)	7 (5 - 10)	1 (1 - 3)
Cool Cucumber	13 (6 - 23)	6 (4 - 7)	2 (2 - 2)
Virginia Tobacco	1 (-)	9 (5 - 14)	1 (-)
Crème Brulee	-	10 (9 - 11)	-
SMOK Infinix (MB) ^a	3 (2 - 3)	14 (3 - 25)	8 (8)
SMOK Infinix (MM) ^b	1 (1 - 1)	-	-
Suorin Edge (MM) ^b	2 (1 - 2)	-	-
JUUL-Jones (CM) ^c	2999 (1766 - 4637)	4573 (3753 - 5174)	79 (60 - 91)
PHIX (HS) ^d	-	156 (46 - 342)	-
KILO 1K (DFI) ^e	-	-	-
KWIT Stick (MT) ^f	61035 (53677 - 66051)	53622 (46134 - 64737)	943 (754 - 1310)

^a MB = Mango Bomb, ^b MM = Mentholate Mango, ^c CM = Clear Mango,

^d HS = Hard Strawberry, ^e DFI = Dewberry Fruit Ice, ^f MT = Mango Tango.

Columns with “ - ” = no data.

Table S5: Mean Concentration (µg/L) of Sum of Other Elements in U-cigarettes and E-cigarettes

Sum of Elements	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Green Mint	1725 (1384 - 2190)	2118 (1128 - 2915)	403 (195 - 757)
Polar Mint	1577 (1266 - 2191)	2370 (1581 - 3021)	446 (253 - 633)
Blueberry Ice	1629 (1436 - 1881)	2378 (1840 - 2674)	292 (128 - 574)
Watermelon Ice	2130 (1472 - 2492)	3252 (2099 - 3951)	379 (74 - 968)
Fruit Medley	64 (54 - 79)	63 (60 - 64)	27 (10 - 38)
Mango	94 (68 - 123)	151 (140 - 159)	80 (28 - 150)
Classic Menthol	68 (60 - 75)	70 (68 - 74)	52 (44 - 56)
Classic Tobacco	90 (81 - 97)	114 (111 - 118)	37 (27 - 44)
Cool Mint	218 (192 - 234)	206 (128 - 258)	117 (64 - 186)
Cool Cucumber	196 (183 - 213)	261 (241 - 283)	189 (132 - 300)
Virginia Tobacco	162 (155 - 176)	182 (154 - 213)	104 (76 - 130)
Crème Brulee	167 (157 - 181)	163 (155 - 176)	83 (51 - 140)
SMOK Infinix (MB) ^a	21 (16 - 29)	230 (146 - 295)	196 (175 - 211)
SMOK Infinix (MM) ^b	279 (267 - 300)	412 (341 - 478)	407 (360 - 432)
Suorin Edge (MM) ^b	282 (270 - 303)	304 (293 - 321)	202 (150 - 252)
JUUL-Jones (CM) ^c	78 (68 - 94)	372 (312 - 428)	21 (7 - 36)
PHIX (HS) ^d	591 (581 - 598)	1750 (1528 - 2165)	249 (182 - 330)
KILO 1K (DFI) ^e	123 (90 - 158)	146 (128 - 157)	62 (53 - 75)
KWIT Stick (MT) ^f	4758 (4078 - 5703)	4847 (3614 - 6344)	107 (63 - 149)

^a MB = Mango Bomb, ^b MM = Mentholate Mango, ^c CM = Clear Mango,

^d HS = Hard Strawberry, ^e DFI = Dewberry Fruit Ice, ^f MT = Mango Tango.

Columns with “ - ” = no data.

Table S6: p-values of Statistical Comparisons of Concentrations of Silicon, Nickel, and the Sum of Other Elements

	Figure 2 - Silicon				Figure 3 - Nickel				Figure 4 – Sum of Elements		
	Unvaped vs. Vaped	Unvaped vs. Aerosol	Vaped vs. Aerosol		Unvaped vs. Vaped	Unvaped vs. Aerosol	Vaped vs. Aerosol		Unvaped vs. Vaped	Unvaped vs. Aerosol	Vaped vs. Aerosol
Green Mint	0.7105	0.0013			*0.5124	-	-		0.9154	0.0159	-
Polar Mint	0.8576	0.0063			*0.5918	-	-		0.3723	0.01	-
Blueberry Ice	0.1618	0.0001			*0.2228	-	-		0.5715	0.0047	-
Watermelon Ice	0.9594	0.0140			*0.2015	-	-		0.7887	0.0254	-
Fruit Medley	0.0394	0.0002			-	-	*0.0008		0.9931	0.0153	-
Mango	0.9076	0.0007			<0.0001	0.9222	-		0.2108	0.8882	-
Classic Menthol	0.6850	0.0011			*0.1958	-	-		0.846	0.0297	-
Classic Tobacco	0.0255	<0.0001			-	-	-		0.0109	0.0002	-
Cool Mint	0.1808	0.0145			0.0121	0.9725	-		0.9561	0.1157	-
Cool Cucumber	0.0532	0.0335			0.2451	0.0773	-		0.3496	0.9842	-
Virginia Tobacco	0.5217	0.0951			-	-	-		0.5286	0.044	-
Crème Brulee	0.6055	0.3695			-	-	-		0.987	0.0256	-
SMOK Infinix (MB) ^a			*0.0008		*0.2026	-	-		0.0025	0.006	-
SMOK Infinix (MM) ^b			*0.1603		-	-	-		0.0238	0.0283	-
Suorin Edge (MM) ^b			*0.2513		-	-	-				-
JUUL-Jones (CM) ^c	0.3383	0.0745			0.1509	0.0168	-		<0.0001	0.158	-
PHIX (HS) ^d	0.3101	0.4353			-	-	-		0.001	0.1617	-
KILO 1K (DFI) ^e	0.1713	0.0014			-	-	-		0.3995	0.0307	-
KWIT Stick (MT) ^f	0.2320	0.9707			0.3664	<0.0001	-		0.9902	0.0016	-

For the statistical analysis, a one-way ANOVA with Dunnett's post hoc test or alternatively data was analyzed using an unpaired t-test. using GraphPad Prism software. *= t-tests

Columns with “-“ = measurement were less than 3 and statistical analysis was not possible.

^aMB = Mango Bomb, ^bMM = Mentholate Mango, ^cCM = Clear Mango, ^dHS = Hard Strawberry, ^eDFI = Dewberry Fruit Ice, ^fMT = Mango Tango

Table S7: Mean Concentrations (µg/L) of Elements in SURGE Mint Flavors

Green Mint	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	-	-	-
Chromium	-	-	-
Cobalt	-	-	-
Copper	375 (314.1 - 477.2)	421 (179.6 - 687.1)	26 (-)
Iron	7 (6.6 - 7.1)	7 (6.6 - 7.1)	30 (-)
Manganese	1 (0.3 - 1.1)	1 (0.4 - 1.4)	1 (0.1 - 2.8)
Silver	2 (-)	2 (0.5 - 2.8)	-
Titanium	4 (3.0 - 5.0)	5 (3.5 - 6.0)	1 (-)
Zinc	541 (510.0 - 603.2)	590 (400.8 - 813.6)	42 (30.5 - 52.8)
Aluminum	23 (17.8 - 31.4)	34 (21.3 - 44.1)	-
Lead	-	-	2 (-)
Tin	49 (-)	43 (41.3 - 44.0)	11 (2.8 - 20.0)
Arsenic	507 (379.3 - 758.7)	648 (399.2 - 779.1)	227 (136.4 - 408.5)
Selenium	251 (113.5 - 511.9)	382 (116.6 - 530.4)	119 (9.5 - 325.9)

Polar Mint	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	1 (-)	0 (-)	-
Chromium	-	-	-
Cobalt	0 (0.2 - 0.3)	0 (-)	-
Copper	507 (235.7 - 717.3)	538 (312.1 - 666.2)	-
Iron	7 (7.1 - 7.1)	7 (7.1 - 7.1)	7 (-)
Manganese	1 (-)	2 (1.4 - 1.6)	1 (-)
Silver	1 (0.1 - 1.6)	2 (1.4 - 1.6)	0 (-)
Titanium	6 (3.3 - 10.1)	8 (4.5 - 12.6)	2 (-)
Zinc	559 (507.7 - 651.7)	839 (710.5 - 919.9)	87 (19.0 - 154.6)
Aluminum	30 (30.4)	37 (32.0 - 41.4)	9 (-)
Lead	-	-	-
Tin	40 (-)	46 (45.9 - 46.4)	14 (1.4 - 21.6)
Arsenic	819 (-)	811 (796.0 - 825.7)	269 (202.5 - 335.1)
Selenium	200 (29.0 - 539.4)	380 (43.7 - 561.6)	99 (9.5 - 259.0)

Columns with “ - ” = no data.

Table S8: Mean Concentration (µg/L) of Elements in SURGE Ice Flavors

Blueberry Ice	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	0 (-)	-	0 (0.2)
Chromium	-	-	-
Cobalt	-	-	-
Copper	244 (163.2 - 318.8)	231 (179.7 - 316.7)	-
Iron	60 (57.4 - 62.5)	64 (55.7 - 74.0)	-
Manganese	1 (0.5 - 1.0)	1 (0.7 - 1.3)	0 (0.0 - 0.5)
Silver	-	2 (1.7 - 2.2)	0 (-)
Titanium	15 (13.5 - 17.0)	18 (16.7 - 20.5)	-
Zinc	623 (461.1 - 734.4)	1102 (1002.7 - 1220.4)	-
Aluminum	32 (20.4 - 53.0)	38 (33.0 - 44.6)	-
Lead	14 (-)	-	14 (-)
Tin	28 (4.9 - 50.7)	35 (1.3 - 54.4)	22 (-)
Arsenic	413 (299.8 - 628.0)	543 (339.8 - 648.6)	187 (127.5 - 291.4)
Selenium	218 (101.3 - 447.7)	345 (108.4 - 463.5)	140 (18.6 - 260.5)

Watermelon Ice	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	1 (-)	0 (-)	-
Chromium	-	-	-
Cobalt	0 (0.2 - 0.3)	0 (-)	-
Copper	507 (235.7 - 717.3)	538 (312.1 - 666.2)	-
Iron	7 (7.1 - 7.1)	7 (7.1 - 7.1)	-
Manganese	1 (-)	2 (1.4 - 1.6)	1 (-)
Silver	1 (0.1 - 1.6)	2 (1.4 - 1.6)	0 (-)
Titanium	6 (3.3 - 10.1)	8 (4.5 - 12.6)	2 (-)
Zinc	559 (507.7 - 651.7)	839 (710.5 - 919.9)	87 (19.0 - 154.6)
Aluminum	30 (30.4)	37 (32.0 - 41.4)	9 (-)
Lead	-	-	-
Tin	40 (-)	46 (45.9 - 46.4)	14 (1.4 - 21.6)
Arsenic	819 (-)	811 (796.0 - 825.7)	269 (202.5 - 335.1)
Selenium	200 (29.0 - 539.4)	380 (43.7 - 561.6)	99 (9.5 - 259.0)

Columns with “ - ” = no data.

Table S9: Mean Concentration (µg/L) of Elements in JUUL Fruit Flavors

Fruit Medley	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	-	-	1 (-)
Chromium	-	-	-
Cobalt	1 (1.0 - 1.7)	2 (1.7 - 2.3)	3 (2.6 - 3.5)
Copper	2 (-)	2 (1.7 - 1.7)	-
Iron	-	7 (7.1 - 7.1)	-
Manganese	-	-	-
Silver	-	-	-
Titanium	2 (1.3 - 1.6)	2 (1.5 - 1.6)	1 (0.6 - 1.5)
Zinc	7 (7.1 - 7.1)	15 (10.2 - 19.0)	-
Aluminum	3 (3.0 - 3.0)	3 (3.0 - 3.0)	15 (-)
Lead	-	-	-
Tin	47 (41.0 - 55.0)	33 (32.1 - 35.0)	18 (7.1 - 33.3)
Arsenic	-	-	-
Selenium	9 (-)	-	-

Mango	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	-	-	1 (-)
Chromium	-	1 (1.4 - 1.4)	-
Cobalt	1 (0.2 - 0.7)	1 (0.5 - 0.6)	1 (1.1 - 1.6)
Copper	2 (1.7 - 1.7)	4 (1.7 - 5.9)	-
Iron	7 (7.1 - 7.1)	22 (17.0 - 31.8)	-
Manganese	-	0 (0.4 - 0.4)	-
Silver	-	-	-
Titanium	1 (1.4 - 1.5)	2 (2.0 - 2.7)	1 (0.6 - 1.4)
Zinc	41 (15.5 - 60.9)	71 (55.0 - 84.4)	-
Aluminum	12 (7.1 - 21.7)	11 (7.1 - 16.2)	60 (9.4 - 127.2)
Lead	-	-	-
Tin	33 (30.1 - 34.6)	40 (34.4 - 46.9)	17 (14.9 - 19.0)
Arsenic	-	-	2 (-)
Selenium	-	-	-

Columns with “ - ” = no data.

Table S10: Mean Concentration (µg/L) of Elements in JUUL Mint/Menthol Flavors

Classic Menthol	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	1 (0.7 - 1.3)	1 (0.9 - 1.4)	5 (4.5 - 4.8)
Chromium	-	-	-
Cobalt	1 (0.9 - 1.8)	1 (1.1 - 1.6)	7 (6.5 - 7.4)
Copper	2 (1.7 - 1.7)	2 (1.7 - 1.7)	15 (8.4 - 18.6)
Iron	15 (13.5 - 16.4)	16 (15.5 - 17.5)	-
Manganese	1 (0.7 - 0.8)	1 (1.0 - 1.1)	-
Silver	2 (1.4 - 1.5)	3 (2.3 - 3.1)	15 (14.3 - 16.7)
Titanium	1 (0.5 - 0.8)	0 (0.3 - 0.5)	0 (0.1 - 0.3)
Zinc	7 (0.4 - 13.6)	7 (7.1 - 7.1)	5 (3.3 - 6.4)
Aluminum	9 (7.3 - 10.8)	13 (12.4 - 13.4)	-
Lead	-	-	-
Tin	30 (28.0 - 30.9)	26 (24.2 - 27.4)	4 (4.2 - 4.2)
Arsenic	-	-	5 (-)
Selenium	-	-	-

Cool Mint	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	1 (0.6 - 0.7)	0 (0.2 - 0.2)	1 (-)
Chromium	-	1 (-)	-
Cobalt	0 (0.1 - 0.2)	0 (0.00.0)	-
Copper	2 (1.7 - 1.7)	2 (1.7 - 1.7)	0 (-)
Iron	13 (11.4 - 13.8)	17 (10.2 - 21.3)	7 (-)
Manganese	0 (0.4 - 0.4)	1 (0.4 - 0.8)	0 (-)
Silver	-	-	-
Titanium	2 (1.6 - 1.7)	2 (1.1 - 2.0)	1 (-)
Zinc	55 (36.5 - 73.4)	63 (29.3 - 91.2)	7 (-)
Aluminum	13 (3.0 - 28.8)	6 (3.0 - 7.1)	15 (5.5 - 34.8)
Lead	-	-	-
Tin	71 (70.9 - 71.3)	63 (46.0 - 74.7)	45 (24.4 - 66.0)
Arsenic	-	-	-
Selenium	61 (58.0 - 66.0)	52 (36.0 - 63.0)	52 (35.0 - 70.0)

Columns with “ - ” = no data.

Table S11: Mean Concentration (µg/L) of Elements in JUUL Tobacco Flavors

Classic Tobacco	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	2 (2.3 - 2.6)	2 (1.9 - 2.1)	1 (1.0 - 1.7)
Chromium	-	-	-
Cobalt	-	-	3 (3.2 - 3.8)
Copper	-	-	9 (4.1 - 11.7)
Iron	14 (12.0 - 18.1)	25 (24.7 - 24.7)	-
Manganese	2 (1.3 - 1.6)	2 (1.7 - 2.1)	-
Silver	-	-	7 (6.5 - 7.5)
Titanium	1 (0.7 - 1.0)	1 (1.0 - 1.0)	0 (-)
Zinc	13 (7.1 - 18.0)	13 (12.5 - 12.7)	-
Aluminum	11 (9.6 - 14.0)	11 (10.8 - 12.1)	-
Lead	-	-	-
Tin	51 (46.1 - 60.3)	51 (47.8 - 54.2)	13 (10.5 - 16.1)
Arsenic	-	-	5 (5.4 - 5.4)
Selenium	-	9 (9.0 - 9.0)	-
Virginia Tobacco	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	1 (0.2 - 0.7)	0 (0.2 - 0.5)	1 (0.4 - 0.6)
Chromium	-	1 (-)	-
Cobalt	-	-	-
Copper	2 (1.7 - 3.3)	3 (1.7 - 4.4)	-
Iron	-	7 (-)	-
Manganese	0 (0.4 - 0.4)	0 (0.4 - 0.4)	-
Silver	-	-	-
Titanium	1 (0.5 - 1.4)	1 (0.5 - 1.0)	-
Zinc	10 (0.4 - 19.2)	17 (7.1 - 37.8)	-
Aluminum	4 (3.0 - 7.1)	4 (3.0 - 7.1)	7 (5.3 - 9.9)
Lead	-	-	-
Tin	67 (64.3 - 68.1)	65 (59.8 - 67.7)	40 (30.2 - 50.7)
Arsenic	-	-	-
Selenium	80 (76.0 - 84.0)	86 (81.0 - 86.0)	55 (40.0 - 69.0)

Columns with “ - ” = no data.

Table S12: Mean Concentration (µg/L) of Elements in Other JUUL Flavors

Cool Cucumber	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	-	-	1 (-)
Chromium	1 (1.4 - 1.4)	5 (4.3 - 5.7)	1 (1.4 - 1.4)
Cobalt	-	-	-
Copper	0 (0.00.4)	1 (0.4 - 2.4)	7 (1.7 - 16.5)
Iron	17 (14.2 - 19.6)	47 (37.9 - 63.0)	#DIV/0!
Manganese	1 (0.8 - 1.1)	1 (1.1 - 1.6)	0 (0.1 - 0.4)
Silver	-	-	-
Titanium	0 (0.2 - 0.2)	1 (0.4 - 0.7)	1 (0.6 - 0.6)
Zinc	5 (-)	7 (-)	91 (23.9 - 202.1)
Aluminum	-	-	18 (4.0 - 25.6)
Lead	-	-	14 (4.7 - 24.5)
Tin	52 (50.3 - 54.7)	60 (56.9 - 62.6)	18 (17.7 - 22.2)
Arsenic	-	-	-
Selenium	122 (116.0 - 131.0)	143 (132.0 - 151.0)	39 (30.0 - 49.0)

Crème Brulee	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	1 (1.1 - 1.4)	1 (1.0 - 1.2)	0 (0.2 - 0.5)
Chromium	-	1 (0.01.4)	-
Cobalt	1 (0.4 - 0.6)	1 (0.4 - 1.1)	1 (0.7 - 1.2)
Copper	2 (1.7 - 1.7)	3 (1.7 - 3.9)	19 (9.3 - 35.1)
Iron	6 (3.9 - 7.2)	13 (9.8 - 17.2)	-
Manganese	0 (0.4 - 0.4)	1 (0.4 - 0.6)	0 (0.1 - 0.4)
Silver	2 (1.4 - 1.9)	1 (0.5 - 1.6)	3 (1.8 - 3.3)
Titanium	1 (0.6 - 1.7)	2 (1.1 - 2.1)	-
Zinc	7 (7.1 - 7.1)	12 (7.1 - 16.3)	54 (-)
Aluminum	13 (12.1 - 14.3)	13 (12.5 - 12.8)	-
Lead	-	-	-
Tin	63 (60.3 - 66.9)	51 (46.1 - 56.8)	18 (15.9 - 20.6)
Arsenic	-	-	5 (0.05.4)
Selenium	74 (71.0 - 79.0)	68 (58.0 - 79.0)	21 (19.0 - 23.0)

Columns with “ - ” = no data.

Table S13: Mean Concentration (µg/L) of Elements in Other Brand Flavors

SMOK Infinix Mango Bomb (MB)	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	-	-	-
Chromium	-	4 (1.1 - 7.2)	2 (0.2 - 4.2)
Cobalt	-	-	-
Copper	-	36 (16.1 - 66.7)	10 (3.2 - 23.0)
Iron	-	15 (7.1 - 21.7)	8 (7.1 - 10.1)
Manganese	-	-	-
Silver	-	-	-
Titanium	-	-	-
Zinc	-	81 (55.0 - 104.8)	56 (47.0 - 61.7)
Aluminum	10 (7.1 - 15.8)	17 (7.1 - 31.8)	9 (1.5 - 14.8)
Lead	-	17 (13.7 - 23.1)	98 (141.8 - 82.7)
Tin	11 (9.4 - 13.2)	60 (11.7 - 85.9)	13 (9.8 - 14.3)
Arsenic	-	-	-
Selenium	-	-	-
SMOK Infinix Mentholate Mango (MM)	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	-	-	-
Chromium	-	40 (-)	43 (-)
Cobalt	-	-	-
Copper	-	20 (14.7 - 23.6)	3 (1.7 - 5.7)
Iron	-	10 (7.1 - 12.4)	-
Manganese	1 (0.9 - 1.0)	1 (0.5 - 1.8)	-
Silver	-	-	-
Titanium	-	1 (0.7 - 1.3)	-
Zinc	7 (-)	43 (30.9 - 57.3)	8 (7.4 - 8.7)
Aluminum	-	-	-
Lead	-	27 (15.6 - 45.9)	30 (16.5 - 52.1)
Tin	22 (16.6 - 28.8)	25 (14.9 - 32.7)	21 (19.0 - 23.9)
Arsenic	-	-	-
Selenium	254 (249.2 - 263.4)	272 (257.0 - 280.0)	333 (317.9 - 342.5)
Suorin Edge Mentholate Mango (MM)	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	-	-	-
Chromium	-	-	2 (1.4 - 2.1)
Cobalt	-	-	1 (0.5 - 0.5)
Copper	-	5 (3.2 - 7.2)	14 (8.3 - 20.1)
Iron	-	22 (20.6 - 25.3)	14 (7.1 - 21.0)
Manganese	1 (0.9 - 1.0)	2 (1.2 - 1.7)	0 (-)
Silver	-	-	-
Titanium	1 (0.5 - 0.5)	2 (1.3 - 1.7)	1 (0.8 - 1.1)
Zinc	7 (-)	38 (28.8 - 46.5)	27 (14.4 - 38.2)
Aluminum	-	-	-
Lead	-	-	-
Tin	25 (19.4 - 31.6)	35 (31.5 - 37.9)	39 (31.8 - 47.0)
Arsenic	-	-	-
Selenium	254 (249.2 - 263.4)	201 (188.8 - 206.9)	111 (42.0 - 150.5)

Columns with “ - ” = no data.

Table S14: Mean Concentration (µg/L) of Elements in Other Brand Flavors (µg/L)

JUUL_Jones Clear Mango	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	-	-	0 (0.2 - 0.5)
Chromium	26 (23.7 - 27.2)	70 (40.9 - 101.5)	-
Cobalt	-	-	-
Copper	7 (5.7 - 10.0)	73 (71.3 - 74.0)	26 (7.3 - 44.6)
Iron	67 (65.6 - 68.4)	334 (245.3 - 440.0)	-
Manganese	1 (1.1 - 1.5)	4 (3.4 - 5.2)	0 (0.2 - 0.3)
Silver	-	12 (-)	-
Titanium	-	7 (3.9 - 9.7)	-
Zinc	240 (231.9 - 246.2)	741 (637.2 - 920.8)	21 (7.1 - 34.8)
Aluminum	27 (23.5 - 31.3)	194 (136.6 - 256.8)	22 (-)
Lead	-	5 (-)	-
Tin	-	92 (60.1 - 117.3)	-
Arsenic	-	-	-
Selenium	222 (218.7 - 225.7)	234 (210.0 - 252.4)	211 (174.1 - 250.2)
PHIX Hard Strawberry (HS)	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	-	-	-
Chromium	1 (0.4 - 1.6)	3 (1.2 - 3.8)	0 (-)
Cobalt	-	-	-
Copper	-	-	6 (-)
Iron	7 (7.1 - 7.1)	13 (7.1 - 16.3)	-
Manganese	-	-	-
Silver	1 (-)	1 (0.5 - 0.6)	-
Titanium	1 (0.7 - 1.5)	7 (2.2 - 12.4)	5 (-)
Zinc	10 (10.0 - 11.0)	19 (14.5 - 24.9)	14 (5.0 - 22.7)
Aluminum	-	8 (-)	-
Lead	-	0 (-)	-
Tin	3 (-)	-	-
Arsenic	-	-	-
Selenium	102 (136.0 - 99.5)	102 (95.3 - 108.3)	49 (40.8 - 59.2)

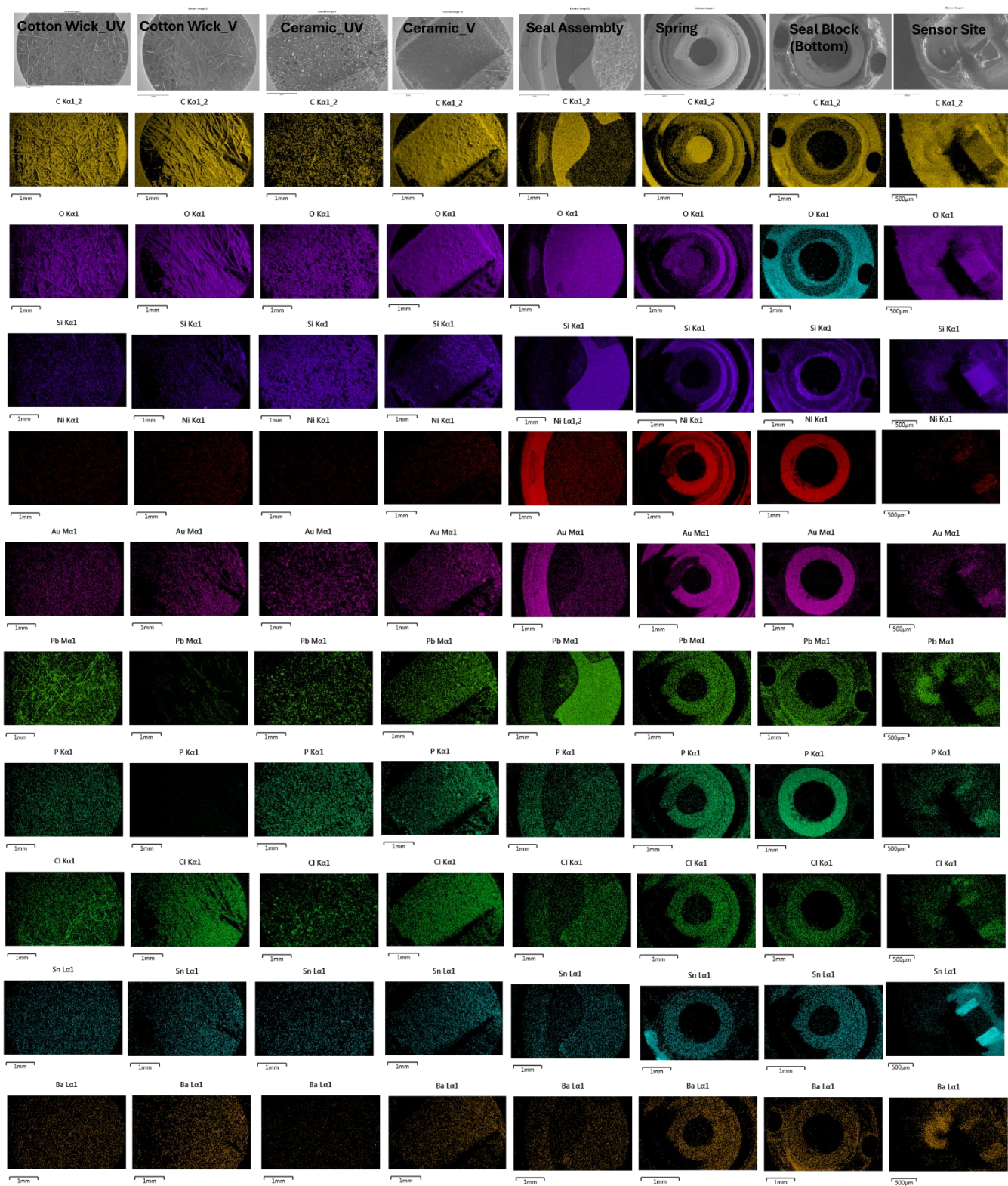
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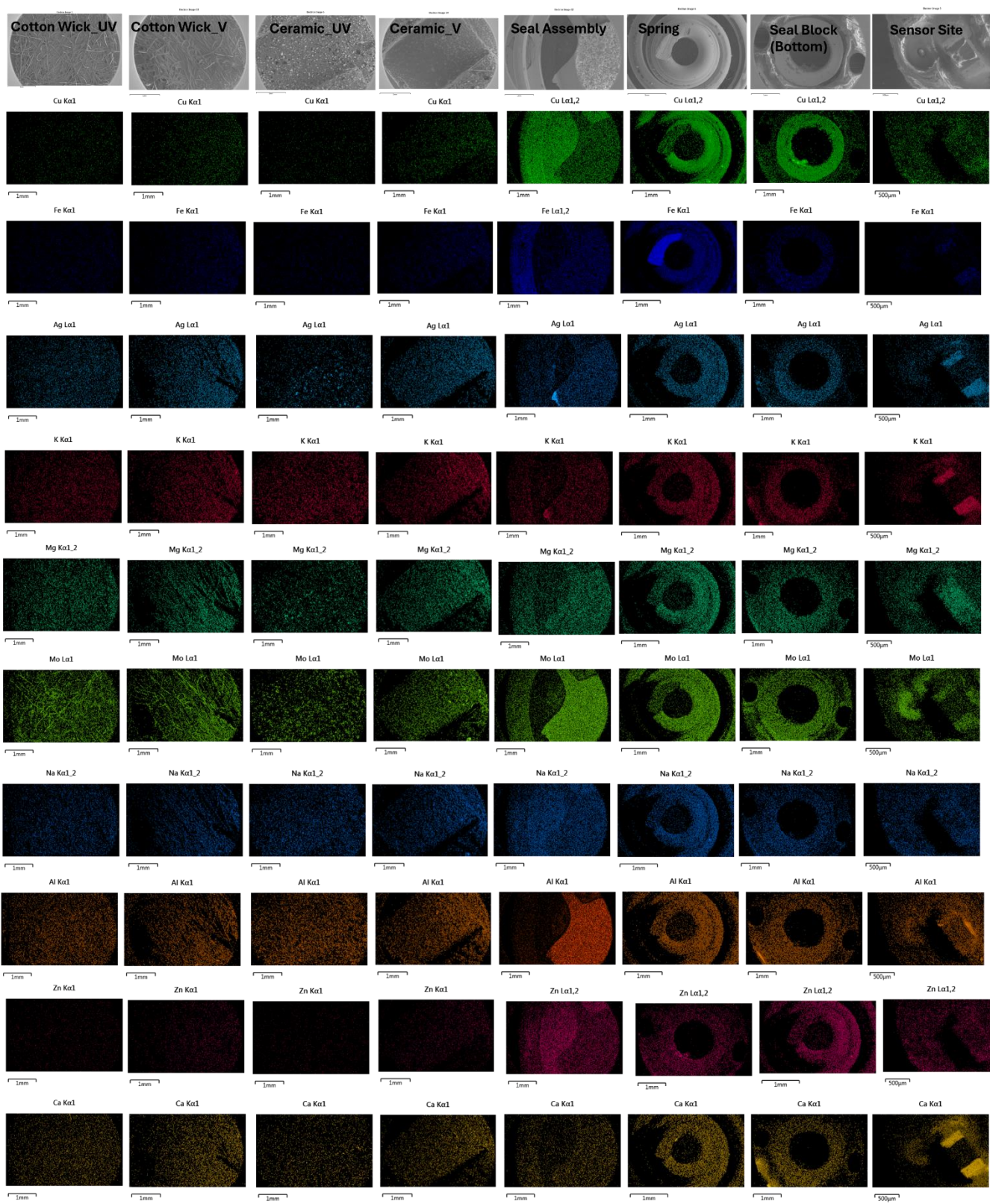
Table S15: Mean Concentration (µg/L) of Elements in Other Brand Flavors

Kilo 1K Dewberry Fruit Ice	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	-	-	-
Chromium	4 (4.0 - 4.9)	10 (6.6 - 14.6)	0 (-)
Cobalt	-	-	-
Copper	13 (-)	104 (100.8 - 107.0)	10 (3.1 - 17.2)
Iron	32 (27.4 - 37.1)	63 (60.4 - 64.5)	-
Manganese	-	1 (0.4 - 0.5)	-
Silver	-	-	-
Titanium	-	-	-
Zinc	11 (7.1 - 19.0)	128 (85.8 - 152.0)	-
Aluminum	19 (15.6 - 22.7)	56 (32.9 - 92.0)	11 (1.7 - 16.6)
Lead	-	2 (-)	1 (0.2 - 2.0)
Tin	7 (7.1 - 7.1)	10 (7.1 - 15.0)	3 (-)
Arsenic	-	-	-
Selenium	-	-	-

KWIT Stick Mango Tango	Unvaped Fluid Mean (Range)	Vaped Fluid Mean (Range)	Aerosol Mean (Range)
Cadmium	-	-	-
Chromium	10 (7.7 - 12.1)	12 (4.6 - 19.2)	-
Cobalt	7 (5.3 - 7.5)	6 (4.5 - 7.0)	-
Copper	2169 (1829.4 - 2615.6)	2153 (1512.1 - 2985.8)	63 (42.0 - 83.9)
Iron	21 (19.9 - 21.0)	25 (20.4 - 32.8)	-
Manganese	24 (4.1 - 41.9)	39 (33.4 - 45.9)	-
Silver	-	-	-
Titanium	-	-	-
Zinc	1856 (1496.4 - 2516.8)	1642 (999.7 - 2547.1)	23 (5.5 - 38.7)
Aluminum	33 (24.7 - 39.8)	38 (20.1 - 64.6)	2 (-)
Lead	607 (434.9 - 815.1)	903 (622.4 - 1090.1)	-
Tin	35 (13.7 - 64.5)	30 (14.5 - 57.4)	21 (13.6 - 26.2)
Arsenic	-	-	-
Selenium	-	-	-

Columns with “ - ” = no data.





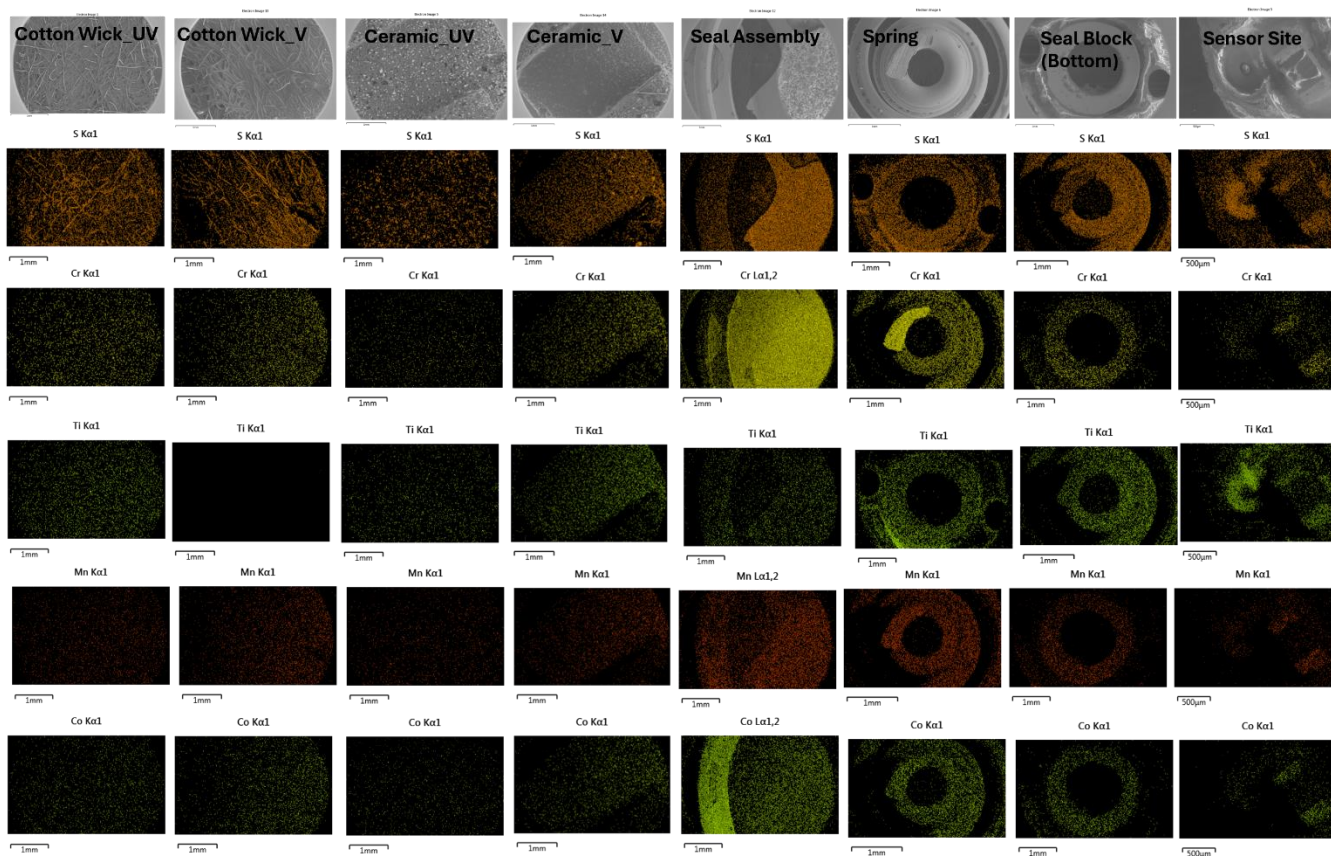


Figure S1: Scanning electron microscopy (SEM) and energy dispersive X-ray spectrometry (EDS) of SURGE components.

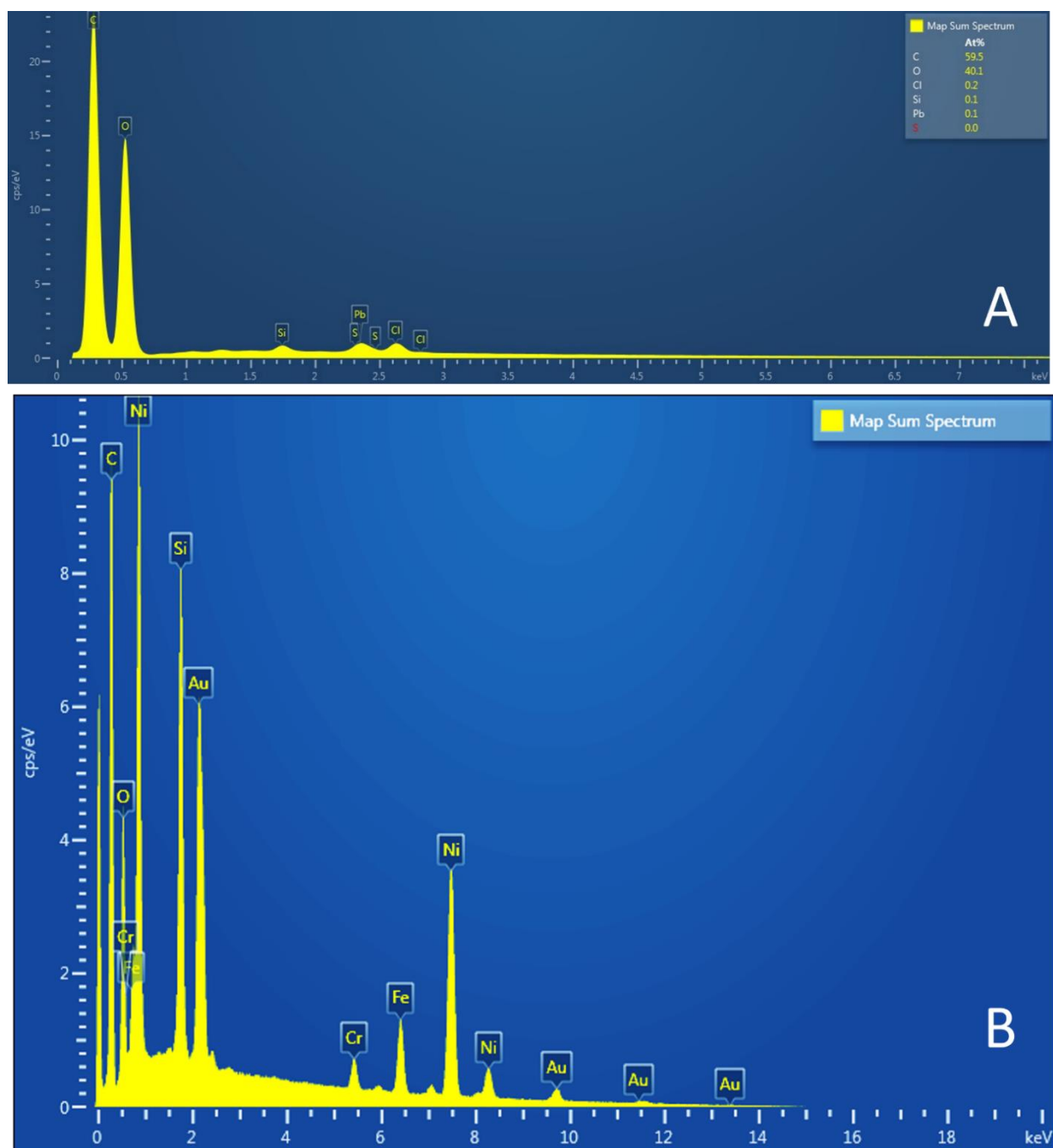


Figure S2: EDS spectra of SURGE sonicator components. (A) vaporized cotton wick, which was mainly carbon, oxygen, silicon and lead. (B) sonicator spring, which was mainly nickel, and gold. Abbreviations: C = carbon; O = oxygen; Cl = chlorine; Si = silicon; Pb = lead; S = sulphur; Cr = chromium; Fe = iron; Au = gold.

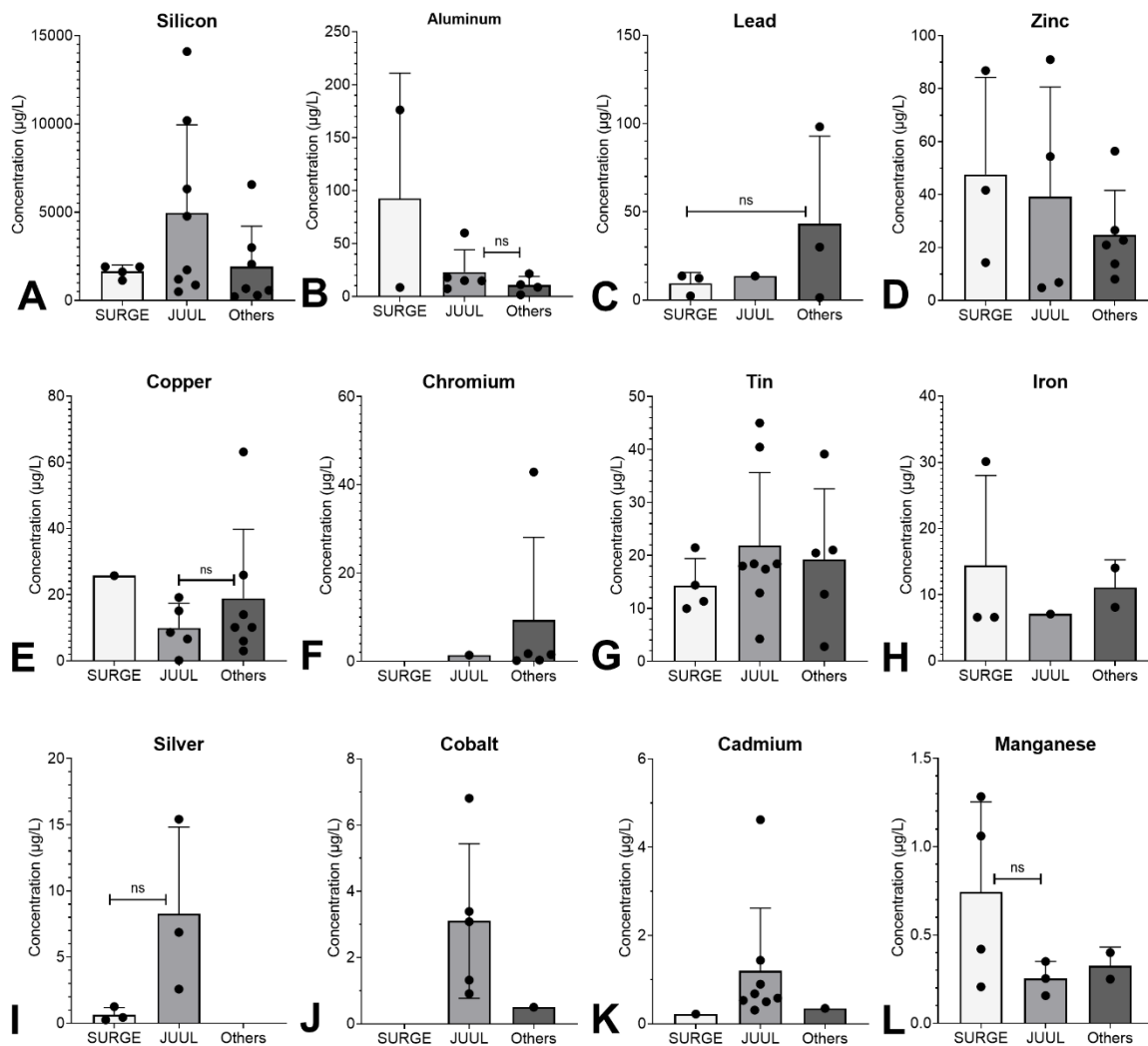


Figure S3: Concentrations of Elements/metals in Aerosols Across Groups. (A) Silicon, (B) Aluminum, (C) Lead, (D) Zinc, (E) Copper, (F) Chromium, (G) Tin, (H) Iron, (I), Silver, (J) Cobalt, (K), Cadmium, (L) Manganese. The x-axis indicates the groups and the y-axis shows concentrations of elements in $\mu\text{g/L}$. Each bar is the mean $\pm$ standard deviation of samples in each group. For comparisons of two groups, an unpaired t-test was used. When comparisons were done for three groups, a one-way ANOVA was used. * = $p < 0.05$; ** = $p < 0.01$; *** = $p < 0.001$; **** = $p < 0.0001$. Brackets are comparisons between groups where indicated.